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Inventor(s)

OTANI; Yoichiro et al.

NEUTRALIZATION PROCESSING SYSTEM FOR AMMONIA-CONTAINING WASTE FLUID AND NEUTRALIZATION PROCESSING METHOD FOR AMMONIA-CONTAINING WASTE FLUID

Abstract

A neutralization processing system for an ammonia-containing waste fluid, includes: one or more sludge tanks; a sludge discharge line configured to supply the waste fluid to the sludge tank; a neutralizer supply line configured to supply a neutralizer to the sludge tank; and a sensor configured to measure pH of the waste fluid or an ammonia concentration of the waste fluid.

Inventors: OTANI; Yoichiro (Kanagawa, JP), MATSUNARI; Kenji (Kanagawa, JP), SAITO; Shunma (Kanagawa, JP)

Applicant: MITSUBISHI KAKOKI KAISHA, LTD. (Kanagawa, JP)

Family ID: 94298172

Assignee: MITSUBISHI KAKOKI KAISHA, LTD. (Kanagawa, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is based upon and claims the benefit of priority from prior Japanese patent application No. 2024-030240 filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field of the Invention

[0002] The present disclosure relates to a neutralization processing system for an ammonia-containing waste fluid and a neutralization processing method for an ammonia-containing waste fluid.

2. Description of the Related Art

[0003] As a countermeasure against global warming, an ammonia fuel, which does not discharge carbon dioxide as a greenhouse gas during combustion, has attracted attention as a promising energy for implementing a carbon-neutral society. In recent years, the practical use of ammonia combustion gas turbines and ammonia combustion engines, which use liquid ammonia as a fuel, has been advanced. In addition, the development for operation of an ammonia fuel ship as a so-called zero-emission ship which uses the liquid ammonia as a fuel also has been advanced.

[0004] However, when the liquid ammonia is used for a fuel of a ship or the like, unburned ammonia gas in the engine is dissolved in moisture in lubricating oil. The moisture is mixed into the lubricating oil due to moisture during the operation or condensation after the engine is stopped. When the engine is operated without removing the moisture, an ammonia concentration in the engine increases, and the ammonia is dissolved in the moisture in the lubricating oil. There is a problem that a material exposed to ammonia water and ammonia atmosphere is likely to be corroded. Therefore, the lubricating oil or the like containing ammonia is discarded as a waste fluid.

[0005] Such an ammonia-containing waste fluid is recovered in a recovery tank or neutralized (see, for example, JP6934555B 1 and JP3863610B).

[0006] Since an installation space of the recovery tank or the like is limited in inboard and land power plant facilities, it is desired that the ammonia-containing waste fluid can be neutralized more efficiently.

SUMMARY

[0007] An object of the present disclosure is to provide a neutralization processing system for an ammonia-containing waste fluid and a neutralization processing method for an ammonia-containing waste fluid, which can solve the above-described problems and efficiently neutralize the ammonia-containing waste fluid.

[0008] According to an aspect of the present disclosure, there is provided a neutralization processing system for an ammonia-containing waste fluid, including: one or more sludge tanks; a sludge discharge line configured to supply the waste fluid to the sludge tank; a neutralizer supply line configured to supply a neutralizer to the sludge tank; and a sensor configured to measure pH of the waste fluid or an ammonia concentration of the waste fluid.

[0009] According to an aspect of the present disclosure, there is provided a neutralization processing system for an ammonia-containing waste fluid, including: a sludge discharge line configured to discharge sludge; one or more waste fluid tanks provided at an end of the sludge

discharge line; a sludge tank configured to store a waste fluid from the waste fluid tank; a waste fluid discharge line configured to supply the waste fluid to the sludge tank; a neutralizer supply line configured to supply a neutralizer to the waste fluid tank; and a sensor configured to measure pH of the waste fluid or an ammonia concentration of the waste fluid in the waste fluid tank.

[0010] According to an aspect of the present disclosure, there is provided a neutralization processing method for an ammonia-containing waste fluid, including: measuring pH or an ammonia concentration of the waste fluid by a sensor; and neutralizing the waste fluid by adding a neutralizer to the waste fluid in accordance with either or both of a measured value of the pH of the waste fluid and a measured value of the ammonia concentration of the waste fluid measured by the sensor.

[0011] The neutralization processing system for an ammonia-containing waste fluid and the neutralization processing method for an ammonia-containing waste fluid of the present disclosure can efficiently neutralize an ammonia-containing waste fluid.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The present disclosure will become more fully understood from the detailed description given hereinbelow and the accompanying drawing which is given by way of illustration only, and thus is not limitative of the present disclosure and wherein:

[0013] FIG. 1 is a block diagram showing a neutralization processing system for an ammonia-containing waste fluid and a neutralization processing method for an ammonia-containing waste fluid according to a first embodiment of the present disclosure;

[0014] FIG. 2 is a block diagram showing a modification of the neutralization processing system for an ammonia-containing waste fluid and the neutralization processing method for an ammonia-containing waste fluid according to the first embodiment of the present disclosure;

[0015] FIG. 3 is a block diagram showing a modification of the neutralization processing system for an ammonia-containing waste fluid and the neutralization processing method for an ammonia-containing waste fluid according to the first embodiment of the present disclosure;

[0016] FIG. 4 is a block diagram showing a neutralization processing system for an ammonia-containing waste fluid and a neutralization processing method for an ammonia-containing waste fluid according to a second embodiment of the present disclosure;

[0017] FIG. 5 is a diagram showing a neutralization processing system for an ammonia-containing waste fluid and a neutralization processing method for an ammonia-containing waste fluid according to a third embodiment of the present disclosure, and a schematic diagram showing a waste fluid tank that stores a waste fluid; and

[0018] FIG. 6 is a diagram showing a neutralization processing system for an ammonia-containing waste fluid and a neutralization processing method for an ammonia-containing waste fluid according to a fourth embodiment of the present disclosure, and a schematic diagram showing a waste fluid tank that stores a waste fluid.

DETAILED DESCRIPTION OF THE INVENTION

First Embodiment

[0019] A neutralization processing system **100** for an ammonia-containing waste fluid and a neutralization processing method for an ammonia-containing waste fluid according to a first embodiment of the present disclosure will be described in detail with reference to FIG. 1.

Neutralization Processing System for Ammonia-Containing Waste Fluid

[0020] The neutralization processing system (hereinafter, appropriately referred to as a “neutralization processing system”) **100** for an ammonia-containing waste fluid shown in FIG. 1 is a device for purifying and neutralizing a waste fluid (hereinafter, referred to as a “waste fluid WF”)

such as a lubricating oil LO, a fuel oil, and bilge water containing ammonia (hereinafter, referred to as “ammonia AM”) used in an engine **200** such as a combustion engine installed in an installation space limited in a ship or land (for example, remote island power plants or existing power plants). [0021] Hereinafter, a case in which the lubricating oil LO used in the engine **200** for [0022] combusting liquid ammonia is treated as the waste fluid WF will be described as an example of the neutralization processing system **100** according to the present disclosure. The neutralization processing system **100** according to the present disclosure can be used for all ammonia waste fluids as long as they are the waste fluid WF containing the ammonia AM, and the following description will be given of an example in which the neutralization processing system **100** includes a purifier **1**. In the engine **200** using the liquid ammonia as a fuel, sludge SG or ammonia water AW may be mixed into the lubricating oil LO in the engine **200**.

[0023] The neutralization processing system **100** includes the purifier **1**. The purifier **1** is a three-phase separation type disk stack centrifuge that separates a stock solution DO which is the lubricating oil LO mixed with the ammonia water (Hereinafter, it is referred to as “ammonia water AW”.) into a light fluid which is the purified lubricating oil LO, a heavy fluid which is the ammonia water AW containing the ammonia AM, and sludge which is a solid content (hereinafter, it is referred to as “sludge SG”). The neutralization processing system **100** mainly includes a lubricating oil tank **300**, a lubricating oil supply pump P1, the purifier **1**, a waste fluid tank **500**, a neutralization tank **600**, a sludge tank **700**, a heavy fluid discharge line **120**, a neutralizer supply line **170**, a waste fluid discharge line **180**, a sensor **720** (a pH meter **721**, an ammonia sensor **722**, and a liquid level measuring instrument **730**), a stock solution supply line **130**, and a sludge discharge line **160**. The neutralization processing system **100** is disposed in the ship.

Engine

[0024] For example, the engine **200** shown in FIG. **1** is a ship diesel engine disposed in the ship, and may be a turbine engine that uses, as a fuel, liquid ammonia having a discharge amount of carbon dioxide as a greenhouse gas of zero. The engine **200** includes a mechanism that connects a crankshaft and a piston with only a connecting rod. In the engine **200**, a lateral pressure, which is a lateral force, is generated on the piston, causing uneven wear in a cylinder. Therefore, the lubricating oil LO for improving the movement of the piston is used in the engine **200**. The engine **200** may co-combust the liquid ammonia and a fossil fuel such as natural gas.

Lubricating Oil Discharge Line

[0025] A lubricating oil discharge line **150** shown in FIG. **1** is a piping path for transferring the lubricating oil LO (stock solution DO) used in the engine **200** to the lubricating oil tank **300**. The lubricating oil discharge line **150** is connected to the engine **200** at one end and to the lubricating oil tank **300** at the other end.

Lubricating Oil Tank and Lubricating Oil

[0026] As shown in FIG. **1**, the lubricating oil tank **300** is a tank for storing the stock solution DO containing the lubricating oil LO used in the engine **200**. The stock solution DO to be supplied from the engine **200** to the lubricating oil tank **300** contains the ammonia AM derived from an ammonia fuel. Therefore, the lubricating oil tank **300** is a tank with a lid that prevents the vaporized ammonia AM from leaking outside the lubricating oil tank **300**.

[0027] The stock solution DO contains moisture mixed in the engine **200**.

Ammonia Water and Water

[0028] The ammonia water AW is an alkaline aqueous solution in which the ammonia AM is dissolved in water. The ammonia water AW has a unique stimulus odor. The ammonia water AW may corrode a fluororubber O-ring or a heavy fluid and light fluid impeller made of copper alloy to be used in the purifier **1**. Therefore, the other parts are preferably made of a corrosion-resistant material.

Stock Solution Supply Line

[0029] The stock solution supply line **130** shown in FIG. **1** is a piping path for supplying the **20**

stock solution DO (lubricating oil LO mixed with the sludge SG and the ammonia water AW) stored in the lubricating oil tank **300** to the purifier **1**. The stock solution supply line **130** includes a heater upstream side line **131** from the lubricating oil tank **300** to the heater **400**, and a heater downstream side line **132** from the heater **400** to the purifier **1**. The heater upstream side line **131** is provided with the lubricating oil supply pump **P1**, and the heater downstream side line **132** is provided with a three-way valve **134**.

Lubricating Oil Supply Pump

[0030] The lubricating oil supply pump **P1** is a pump for transferring the stock solution DO in the lubricating oil tank **300** from the stock solution supply line **130** to the purifier **1** via the heater **400**. As shown in FIG. **1**, the ammonia water AW is mixed into the stock solution DO in **30** the heater upstream side line **131**.

Heater

[0031] The heater **400** is an oil heater that heats the lubricating oil LO to be transferred to the purifier **1**. The heater **400** can increase a solid content and/or moisture separation efficiency in the purifier **1** by heating the lubricating oil LO and decreasing a viscosity. As shown in FIG. **1**, the ammonia water AW and the sludge SG are mixed into the stock solution DO in the heater downstream side line **132**.

Light Fluid Discharge Line

[0032] A light fluid discharge line **140** is a flow path for returning the lubricating oil LO (light fluid) purified by the purifier **1** to the lubricating oil tank **300**. The light fluid discharge line **140** is a piping path from a discharge port of the purifier **1** to a lubricating oil supply port of the lubricating oil tank **300**.

Purifier

[0033] The purifier **1** is a device for separating and removing, from the lubricating oil LO, the sludge SG (solid content) or the ammonia water AW contained in the stock solution DO supplied from the lubricating oil tank **300**. The purifier **1** may be, for example, a disk stack centrifuge that separates the stock solution DO into three phases of fluid (light fluid)-fluid (heavy fluid)-solid content by a centrifugal force of a rotating body that rotates at a high speed.

Sludge Discharge Line

[0034] The sludge discharge line **160** is a discharge path for discharging the sludge SG centrifuged by the purifier **1** to the outside of the purifier **1**. The sludge discharge line **160** has an upstream side connected to a sludge discharge port of the purifier **1** and a downstream side connected to the sludge tank **700** via the waste fluid discharge line **180** (see FIG. **1**). The sludge SG in the purifier **1** is discharged to the waste fluid discharge line **180**, stored in the sludge tank **700**, and then discarded.

Waste Fluid Discharge Line

[0035] The waste fluid discharge line **180** is a waste fluid discharge unit that supplies the waste fluid WF to the sludge tank **700**.

Waste Fluid Tank

[0036] The waste fluid tank **500** shown in FIG. **1** is a tank for storing the waste fluid WF containing the sludge SG separated by the purifier **1**. The waste fluid tank **500** is disposed downstream of the sludge discharge line **160**. The number of waste fluid tanks **500** may be one or more.

[0037] The sludge SG is deposited on an outermost diameter portion side in a rotating body **3** by a centrifugal force of the purifier **1**, and is discharged from the sludge discharge port to the sludge discharge line **160**.

[0038] In the first embodiment, the ammonia water AW (heavy fluid) separated by the purifier **1** is also stored in the waste fluid tank **500**.

[0039] Similar to the sludge tank **700**, the waste fluid tank **500** may be provided with an input water supply line **712** and a vent line **770**.

Neutralization Tank, Neutralizer Supply Line, and Waste Fluid

[0040] The neutralization tank **600** is a tank for storing neutralizers (Hereinafter, it is referred to as “neutralizer N”.) that neutralize a mixture (hereinafter, referred to as a waste fluid WF) of the sludge SG and the ammonia water AW stored in the waste fluid tank **500**. The neutralization tank **600** is provided with the neutralizer supply line **170** that supplies the neutralizer N in the neutralization tank **600** to the waste fluid tank **500**.

[0041] The neutralizer supply line **170** is a neutralizer supply unit that supplies the neutralizer N to the waste fluid tank **500**.

Neutralizer

[0042] The neutralizer N is used for neutralizing the waste fluid WF containing the ammonia water AW in the waste fluid tank **500**, and for example, is made of a material having an acidic component such as citric acid. The neutralizer N may be a solid (powder (including a crystalline state)) or a liquid as long as it can neutralize the waste fluid WF containing the ammonia water AW. The neutralizer N is supplied from the neutralizer supply line **170** to the waste fluid WF in the sludge tank **700**. The neutralizer N is supplied in accordance with a measured value of pH, a measured value of an ammonia concentration of the waste fluid WF, and the like detected by the sensor **720**, and neutralize the waste fluid WF. The neutralizer N is preferably citric acid. Citric acid is likely to be handled and has high safety. The neutralizer N may be citric acid powders (crystals of anhydrous citric acid (C.sub.6H.sub.8O.sub.7) or citric acid hydrate (C.sub.6H.sub.8O.sub.7 H.sub.2O)) or an aqueous citric acid solution.

[0043] When the citric acid is used as the neutralizer N, iron oxide may be added to the waste fluid WF as an additive for detoxification. When the citric acid and the iron oxide are added to the ammonia water AW, ferric ammonium citrate is generated. The neutralizer N and the additive for detoxification may be a solid or a liquid.

Sludge Tank

[0044] The sludge tank **700** is a tank for storing the waste fluid WF discharged from the waste fluid tank **500**. The sludge tank **700** is provided with the waste fluid discharge line **180**. When the neutralized waste fluid WF in the sludge tank **700** is 40° C. or higher, a cooling device that cools the waste fluid WF may be disposed.

[0045] The sludge tank **700** that stores the neutralized waste fluid WF obtained by neutralizing and detoxifying the ammonia AM can be easily carried out and moved from an installation place.

Input Water Supply Line

[0046] The input water supply line **712** is a flow path for supplying an appropriate amount of water to the sludge tank **700** or the waste fluid tank **500** in accordance with components, a concentration, or the like of the waste fluid WF in the sludge tank **700** or the waste fluid tank **500**. The input water supply line **712** may be provided as necessary, and may be omitted.

Vent Line

[0047] The vent line **770** is a pipe for discharging gas or water vapor in a gas phase of the sludge tank **700** or the waste fluid tank **500**. The vent line **770** serves to adjust the expansion or contraction of gas caused by a change in a pressure or a temperature of the gas in the sludge tank **700** and to remove unnecessary gas. The vent line **770** may be provided with a valve, a pump, a filter, and the like.

Sensor

[0048] The sensor **720** shown in FIG. **1** or **4** is a detector including the pH meter **721** that measures the pH of the waste fluid WF in the sludge tank **700** or the waste fluid tank **500**, the ammonia sensor **722** that measures the ammonia concentration of the waste fluid WF, and the like.

[0049] The pH meter **721** can efficiently neutralize the waste fluid WF by measuring the pH of the waste fluid WF in the sludge tank **700** or the waste fluid tank **500** and adding an optimum amount of neutralizer N to the waste fluid WF in accordance with the measured value of the pH.

[0050] The ammonia sensor **722** can efficiently neutralize the waste fluid WF by measuring the

ammonia concentration of the waste fluid WF in the sludge tank **700** or the waste fluid tank **500** and adding an optimum amount of neutralizer N to the waste fluid WF in accordance with the measured value of the ammonia concentration.

Heavy Fluid Discharge Line

[0051] The heavy fluid discharge line **120** is a flow path through which the ammonia water AW (separated water) centrifuged by the purifier **1** is discharged. The ammonia water AW in the purifier **1** is stored in the sludge tank **700** and then discarded.

Operations

[0052] Next, operations of the neutralization processing system **100** for an ammonia-containing waste fluid and the neutralization processing method for an ammonia-containing waste fluid according to the first embodiment of the present disclosure will be described with reference to FIG. **1**.

[0053] As shown in FIG. **1**, the lubricating oil LO (stock solution DO) used in the engine **200** is stored in the lubricating oil tank **300**, and then introduced into the purifier **1** from the stock solution supply line **130**.

[0054] The ammonia water AW and the sludge SG (waste fluid WF) separated by the purifier **1** shown in FIG. **1** are discharged from the sludge discharge line **160** to the waste fluid tank **500**, and then transferred from the waste fluid discharge line **180** to the sludge tank **700** and stored therein.

[0055] On the ship, the sludge SG needs to be unloaded and treated as industrial waste. However, when the ammonia concentration of the sludge SG is high, the unloading may be refused. Therefore, it is preferred to neutralize the waste fluid WF containing the ammonia AM in the ship.

[0056] In the waste fluid tank **500**, as shown in FIG. **1**, at least one of the pH or the ammonia concentration of the waste fluid WF stored in the waste fluid tank **500** is measured by the sensor **720** (the pH meter **721** and the ammonia sensor **722**) (measurement step).

[0057] Then, the neutralizer N is added to the waste fluid WF to neutralize the waste fluid WF in accordance with at least one of the measured value of the pH of the waste fluid WF or the measured value of the ammonia concentration of the waste fluid WF measured by the sensor **720** (neutralization step). In this way, the waste fluid WF containing the ammonia AM can be efficiently neutralized.

[0058] As shown in FIG. **1**, the neutralization processing system **100** according to the first embodiment neutralizes the waste fluid WF containing the ammonia AM. The neutralization processing system **100** includes one or more waste fluid tanks **500**, the sludge discharge line **160** that supplies the waste fluid WF to the waste fluid tank **500**, the neutralizer supply line **170** that supplies the neutralizer N to the waste fluid tank **500**, and the sensor **720** (the pH meter **721** and the ammonia sensor **722**) that measures the pH of the waste fluid WF or the ammonia concentration of the waste fluid WF.

[0059] According to such a configuration, the neutralizer N can be supplied to the waste fluid WF in accordance with at least one of the measured value of the pH of the waste fluid WF or the measured value of the ammonia concentration of the waste fluid WF detected by the sensor **720**, and thus the waste fluid WF can be efficiently neutralized.

[0060] Processing of supplying the neutralizer N to the waste fluid WF in accordance with the measured value of the ammonia concentration of the waste fluid WF may be performed by automatically operating a pump or a valve using a control device, or may be performed manually.

[0061] The neutralizer N has an acidic component. If the neutralizer N is acidic, the neutralizer N can neutralize the waste fluid WF containing the ammonia AM.

[0062] The sensor **720** shown in FIG. **1** is either or both of the pH meter **721** and the ammonia sensor **722**. In such a configuration, the pH value or the ammonia concentration can be directly measured, and thus an appropriate amount of neutralizer N can be supplied to the waste fluid WF.

[0063] The neutralizer N preferably contains citric acid.

[0064] According to such a configuration, the citric acid is likely to be handled and has high safety,

and thus is easily carried in and stored.

[0065] As shown in FIG. 1, the waste fluid tank **500** is provided with the input water supply line **712** for supplying water to the waste fluid tank **500**.

[0066] According to such a configuration, the waste fluid tank **500** includes the input water supply line **712**, and thus the waste fluid WF in the waste fluid tank **500** can be appropriately cooled and diluted by supplying water from the input water supply line **712**.

[0067] The neutralization processing system **100** includes one or more waste fluid tanks **500** that store the waste fluid WF.

[0068] According to such a configuration, a neutralization processing system **100A** includes one or more waste fluid tanks **500**, and thus can appropriately store the waste fluid WF in accordance with an amount of the waste fluid WF generated in the ship or the like.

[0069] As shown in FIG. 1, the neutralization processing method for an ammonia-containing waste fluid of the present embodiment neutralizes the waste fluid WF containing the ammonia AM. Specifically, the pH or the ammonia concentration of the waste fluid WF stored in the sludge tank **700** or the waste fluid tank **500** is measured by the sensor **720** (the pH meter **721**, the ammonia sensor **722**, and the like), and the waste fluid WF is neutralized by adding the neutralizer N to the waste fluid WF in accordance with either or both of the measured value of the pH of the waste fluid WF and the measured value of the ammonia concentration of the waste fluid WF measured by the sensor **720**.

[0070] According to such a neutralization processing method for an ammonia-containing waste fluid, the waste fluid WF can be efficiently neutralized by adding the neutralizer N to the waste fluid WF in accordance with the measured value of the pH or the measured value of the ammonia concentration of the waste fluid WF measured by the sensor **720**.

Modification of First Embodiment

[0071] The present disclosure is not limited to the first embodiment, various modifications and changes can be made within the scope of the technical idea of the present disclosure, and it goes without saying that the present disclosure also covers the modified and changed invention.

[0072] Hereinafter, a modification of the neutralization processing system **100** for an ammonia-containing waste fluid and the neutralization processing method for an ammonia-containing waste fluid according to the first embodiment will be described with a focus on differences from the neutralization processing system **100** for an ammonia-containing waste fluid and the neutralization processing method for an ammonia-containing waste fluid according to the first embodiment. The configurations already described are denoted by the same reference numerals, and the description thereof is omitted.

[0073] FIGS. 2 and 3 are block diagrams showing a modification of the neutralization processing system **100** for an ammonia-containing waste fluid and the neutralization processing method for an ammonia-containing waste fluid according to the first embodiment of the present disclosure.

[0074] The neutralization processing system **100** for an ammonia-containing waste fluid shown in FIG. 1 may further include an additive for detoxification tank **780** that stores additives for detoxification (hereinafter, referred to as an “additive for detoxification AD”) and an additive for detoxification supply line **711** which supplies the additive for detoxification AD stored in the additive for detoxification tank **780** to the sludge tank **700** or the waste fluid tank **500**, as in neutralization processing systems **100A** and **100B** for an ammonia-containing waste fluid shown in FIGS. 2 and 3.

[0075] According to such a configuration, the neutralization processing systems **100A** and **100B** include the additive for detoxification tank **780** and the additive for detoxification supply line **711**, and thus the waste fluid WF can be detoxified by supplying the additive for detoxification AD to the waste fluid WF in the sludge tank **700** or the waste fluid tank **500**.

[0076] In this way, the neutralization processing systems **100A** and **100B** for an ammonia-containing waste fluid include the sludge discharge line **160** configured to discharge the sludge

[0077] SG, one or more waste fluid tanks **500** provided at an end of the sludge discharge line **160**, the sludge tank **700** configured to store the waste fluid WF from the waste fluid tank **500**, the waste fluid discharge line **180** configured to supply the waste fluid WF to the sludge tank **700**, the neutralizer supply line **170** configured to supply the neutralizer N to the waste fluid tank **500**, and the sensor **720** configured to measure the pH of the waste fluid WF or the ammonia concentration of the waste fluid WF in the waste fluid tank **500**.

[0078] According to such a configuration, the neutralization processing systems **100A** and **100B** include the sensor **720** that measures the pH of the waste fluid WF or the ammonia concentration of the waste fluid WF in the waste fluid tank **500**, and thus the value of the pH or the ammonia concentration can be directly measured, and thus an appropriate amount of neutralizer N can be supplied to the waste fluid tank **500**.

[0079] As shown in FIG. 2, the neutralization processing system **100A** includes the additive for detoxification supply line **711** which supplies the additive for detoxification AD containing iron oxide to the sludge tank **700**. When the citric acid is used as the neutralizer N, the iron oxide is preferably supplied to the waste fluid WF as an additive for detoxification.

[0080] According to such a configuration, when the citric acid and the iron oxide are added to the ammonia water AW, ferric ammonium citrate is generated, and thus the waste fluid WF can be detoxified.

[0081] As shown in FIG. 3, the neutralization processing system **100B** includes the additive for detoxification supply line **711** which supplies the additive for detoxification AD containing iron oxide to the waste fluid tank **500**. In this case, an upstream side of the additive for detoxification supply line **711** is preferably connected to the additive for detoxification tank **780** that stores the additives for detoxification AD.

[0082] According to such a configuration, the neutralization processing system **100B** includes the additive for detoxification supply line **711**, and thus the waste fluid WF in the waste fluid tank **500** can be detoxified by supplying the additive for detoxification AD to the waste fluid WF.

Second Embodiment

[0083] FIG. 4 is a block diagram showing a neutralization processing system **100C** for an ammonia-containing waste fluid and a neutralization processing method for an ammonia-containing waste fluid according to a second embodiment of the present disclosure.

[0084] The waste fluid tank **500** and the waste fluid discharge line **180** of the first embodiment shown in FIG. 1 may be omitted as in the neutralization processing system **100C** shown in FIG. 4.

[0085] In this case, the downstream side of the sludge discharge line **160** connected to the purifier **1** is connected to the sludge tank **700**. A downstream side of the neutralizer supply line **170** connected to the neutralization tank **600** is connected to the sludge tank **700**. The sludge discharge line **160** and the purifier **1** are not limited to one, and may be plural.

[0086] Similar to the waste fluid tank **500** shown in FIG. 1, the sludge tank **700** is provided with the sensor **720** (the pH meter **721**, the ammonia sensor **722**, and the like).

[0087] In this case, the neutralization processing system **100C** also has the same operation and effect as the neutralization processing system **100** of the first embodiment.

[0088] As shown in FIG. 4, the sludge tank **700** of the neutralization processing system **100C** may be provided with the input water supply line **712** for supplying water to the sludge tank **700**. In this case, the sludge tank **700** is preferably provided with the vent line **770**.

[0089] According to such a configuration, the sludge tank **700** is provided with the input water supply line **712**, and thus the waste fluid WF in the sludge tank **700** can be diluted.

[0090] The sludge tank **700** of the neutralization processing system **100C** may be provided with the additive for detoxification supply line **711** which supplies the additive for detoxification AD, the liquid level measuring instrument **730** for monitoring an amount (liquid level) of the waste fluid WF stored in the sludge tank **700**, and a device or structure **740** having a stirring function to be described later.

Third Embodiment

[0091] FIG. 5 is a diagram showing a neutralization processing system **100D** for an ammonia-containing waste fluid and a neutralization processing method for an ammonia-containing waste fluid according to a third embodiment of the present disclosure, and a schematic diagram showing the sludge tank **700** that stores the waste fluid WF.

[0092] As shown in FIG. 5, the waste fluid tank **500** may be provided with the waste fluid discharge line **180**, a waste fluid transfer pump **761**, the device or structure **740** having a stirring function, the neutralizer supply line **170**, the sludge discharge line **160**, the additive for detoxification supply line **711**, the input water supply line **712**, the vent line **770**, the liquid level measuring instrument **730**, and the sensor **720** (for example, the pH meter **721** and the ammonia sensor **722**).

[0093] A plurality of purifier **1** are provided unlike FIG. 1.

[0094] The waste fluid discharge line **180** is a flow path for transferring the waste fluid WF in the waste fluid tank **500** to the sludge tank **700**. The waste fluid discharge line **180** is provided with the waste fluid transfer pump **761**. The waste fluid transfer pump **761** is a pump for transferring the waste fluid WF in the waste fluid tank **500** to the sludge tank **700**. In the present embodiment, the amount of waste fluid WF stored in the waste fluid tank **500** is measured by the liquid level measuring instrument **730**, and a flow rate of the waste fluid discharge line **180** (a discharge rate of the waste fluid transfer pump **761**) is controlled in accordance with the amount of stored waste fluid WF.

[0095] The sludge tank **700**, the waste fluid discharge line **180**, and the waste fluid transfer pump **761** may be arranged in plural in accordance with the amount of the waste fluid WF to be generated in the ship. In addition, for example, the waste fluid WF can be transferred to any tank in accordance with a liquid level in the tank measured by the liquid level measuring instrument **730**. Alternatively, for example, the waste fluid WF that overflows from the waste fluid tank **500** may be guided to the sludge tank **700**. In this case, the waste fluid transfer pump **761** may be omitted.

[0096] The neutralization processing system **100D** may include one or more sludge tanks **700** that store the neutralized waste fluid WF.

[0097] According to such a configuration, the neutralization processing system **100D** includes one or more sludge tanks **700**, and thus can appropriately store the waste fluid WF in accordance with the amount of the waste fluid WF generated in the ship or the like.

[0098] The waste fluid tank **500** may be provided with the device or structure **740** having a stirring function of stirring the waste fluid WF. The device or structure **740** having a stirring function may be a device or structure having a function of stirring the waste fluid WF in the waste fluid tank **500**, and may be, for example, a stirrer. The device or structure **740** having a stirring function may be appropriately provided in accordance with conditions of the waste fluid WF in the waste fluid tank **500**, and the neutralizer N, the additive for detoxification AD, and the like to be added to the waste fluid WF, and may be omitted.

[0099] According to such a configuration, by providing the device or structure **740** having a stirring function in the waste fluid tank **500**, the neutralizer N or the additive for detoxification AD can be efficiently mixed with the waste fluid WF in the waste fluid tank **500**, and the waste fluid WF can be detoxified.

[0100] Similarly, the sludge tank **700** may be provided with the device or structure **740** having a stirring function of stirring the waste fluid WF.

[0101] According to such a configuration, by providing the device or structure **740** having a stirring function in the sludge tank **700**, the neutralizer N or the additive for detoxification AD can be efficiently mixed with the waste fluid WF in the sludge tank **700**, and the waste fluid WF can be detoxified.

[0102] Similar to the first embodiment, the neutralizer supply line **170** is a conveyance path for supplying the neutralizer N to the waste fluid tank **500**. In the neutralizer supply line **170**, the

neutralizer N in an amount corresponding to the measured value of the pH of the waste fluid WF measured by the sensor **720** (for example, the pH meter **721** and the ammonia sensor **722**) is transferred to the waste fluid tank **500**.

[0103] Similar to the first embodiment, the sludge discharge line **160** is a flow path for discharging the waste fluid WF discharged from the purifier **1** into the waste fluid tank **500**. The sludge discharge line **160** and the purifier **1** are not limited to one, and may be plural. Since the neutralization processing system **100D** is provided with an appropriate number of sludge discharge lines **160** and purifiers **1**, a large amount of waste fluid WF can be suitably treated.

[0104] The additive for detoxification supply line **711** is a conveyance path for supplying the additive for detoxification AD that detoxifies the waste fluid WF in the waste fluid tank **500** to the waste fluid tank **500**. The additive for detoxification can be appropriately selected according to the components of the waste fluid WF. When the neutralizer N is citric acid, the additive for detoxification is preferably iron oxide. In the additive for detoxification supply line **711**, the additive for detoxification AD is transferred to the waste fluid tank **500** in an amount corresponding to the measured value of the pH or the measured value of the ammonia concentration of the waste fluid WF measured by the sensor **720**. The neutralization processing system **100D** can detoxify the waste fluid WF in the waste fluid tank **500** by providing the additive for detoxification supply line **711** and adding the additive for detoxification AD.

[0105] The additive for detoxification supply line **711** may be appropriately provided according to the components of the waste fluid WF in the waste fluid tank **500**, and may be omitted.

[0106] The input water supply line **712** is a flow path for supplying an appropriate amount of water W in accordance with the components, the concentration, and the like of the waste fluid WF in the waste fluid tank **500**. The neutralization processing system **100D** can dilute the waste fluid WF in the waste fluid tank **500** by providing the input water supply line **712** to input water.

[0107] The input water supply line **712** may be appropriately provided according to the components or the conditions of the waste fluid WF in the waste fluid tank **500**, and may be omitted.

[0108] The vent line **770** is a pipe for discharging gas or water vapor in the gas phase of the waste fluid tank **500**. The vent line **770** serves to adjust the expansion or contraction of gas caused by a change in a pressure or a temperature of the gas in the waste fluid tank **500** and to remove unnecessary gas. The vent line **770** may be provided with a valve, a pump, a filter, and the like.

[0109] The liquid level measuring instrument **730** is a meter for measuring the amount (liquid level) of waste fluid WF stored in the waste fluid tank **500**. By providing the liquid level measuring instrument **730**, the waste fluid transfer pump **761** can be driven in accordance with the amount of waste fluid stored in the waste fluid tank **500** to transfer the waste fluid WF to the sludge tank **700**.

[0110] The liquid level measuring instrument **730** may be appropriately provided as necessary, and may be omitted.

[0111] Further, the neutralization processing system **100D** may be provided with a weir instead of the liquid level measuring instrument **730** to allow the waste fluid WF in the waste fluid tank **500** to flow (overflow) from the waste fluid discharge line **180** to the sludge tank **700** when the waste fluid WF in the waste fluid tank **500** reaches a predetermined liquid level.

[0112] Similar to the first embodiment, the sensor **720** includes the pH meter **721**.

[0113] In this way, in the neutralization processing system **100D** for an ammonia-containing waste fluid, the waste fluid tank **500** is provided with the liquid level measuring instrument **730**, and the neutralized waste fluid WF in the waste fluid tank **500** is freely transferred to the sludge tank **700** in accordance with the liquid level of the neutralized waste fluid WF.

[0114] According to such a configuration, the neutralization processing system **100D** can transfer the neutralized waste fluid WF in the waste fluid tank **500** to the sludge tank **700** in accordance with the liquid level of the waste fluid WF. Therefore, if the amount of waste fluid WF stored in the waste fluid tank **500** increases, the waste fluid WF can be automatically transferred to the sludge

tank **700** as appropriate, and a large amount of waste fluid WF can be efficiently stored.

Fourth Embodiment

[0115] FIG. **6** is a diagram showing a neutralization processing system **100E** for an ammonia-containing waste fluid and a neutralization processing method for an ammonia-containing waste fluid according to a fourth embodiment, and a schematic diagram showing the sludge tank **700** that stores the waste fluid WF.

[0116] The case in which the neutralization processing system **100D** for an ammonia-containing waste fluid of the third embodiment (see FIG. **5**) includes one waste fluid tank **500** have been described, and as in the neutralization processing system **100E** shown in FIG. **6**, the neutralization processing system **100D** may include a plurality of waste fluid tanks **500A** and **500B**.

[0117] The neutralization processing system **100E** includes a plurality of waste fluid tanks **500A** and **500B** and one sludge tank **700**. Further, the amount of waste fluid WF that can be stored can be increased, and thus the neutralization processing system **100E** can cope with large ships or large facilities. Further, the neutralization processing can be performed at a plurality of locations, and thus when one of the waste fluid tanks **500A** and **500B** cannot be used due to a maintenance, a failure, or the like, the processing of the waste fluid WF can also be continued.

[0118] It is preferred that the two waste fluid tanks **500A** and **500B** each are provided with the waste fluid discharge line **180**, waste fluid transfer pumps **761A** and **761B**, the device or structure **740** having a stirring function, the neutralizer supply line **170**, the sludge discharge line **160**, the additive for detoxification supply line **711**, the input water supply line **712**, the vent line **770**, the liquid level measuring instrument **730**, and the sensor **720** (for example, the pH meter **721** and the ammonia sensor **722**).

[0119] The waste fluid WF accumulated in the waste fluid tanks **500A** and **500B** may be transferred from the waste fluid discharge line **180** to the sludge tank **700** by driving the waste fluid transfer pumps **761A** and **761B** in accordance with liquid levels in the waste fluid tanks **500A** and **500B**.

Modification 1

[0120] In the first embodiment, the case in which the purifier **1** is the three-phase separation type disk stack centrifuge has been exemplified, and a two-phase separation type disk stack centrifuge may be used as the purifier **1**. When the two-phase separation type disk stack centrifuge is used as the purifier **1**, it is preferred to provide a pressure sensor **840** that detects a pressure of the ammonia water AW to be discharged from a heavy fluid extraction unit **15** of the purifier **1** and supply water to a separation chamber of the purifier **1** when a measured value of the pressure sensor **840** falls below a predetermined value.

Modification 2

[0121] In the first to fourth embodiments and Modification 1, the case in which the neutralization processing systems **100**, **100A**, **100B**, **100C**, **100D**, and **100E** include the purifier **1** is exemplified, and these neutralization processing systems may not include the purifier **1**. In this case, the same operation and effect as those of the neutralization processing systems **100** to **100E** of the first to fourth embodiments and Modification 1 can also be achieved.

Claims

1. A neutralization processing system for an ammonia-containing waste fluid, the neutralization processing system separating a waste fluid containing at least sludge and ammonia from lubricating oil after use in a ship diesel engine to neutralize the waste fluid, the neutralization processing system comprising: a purifier configured to purify the lubricating oil; a sludge tank; a sludge discharge line configured to supply the waste fluid separated from the lubricating oil by the purifier to the sludge tank; a neutralizer supply line configured to supply a neutralizer to the sludge tank; and a sensor configured to measure pH of the waste fluid or an ammonia concentration of the waste fluid.

2. A neutralization processing system for an ammonia-containing waste fluid, the neutralization processing system separating a waste fluid containing at least sludge and ammonia from lubricating oil after use in a ship diesel engine to neutralize the waste fluid, the neutralization processing system comprising: a purifier configured to purify the lubricating oil; a waste fluid tank; a sludge discharge line configured to supply the waste fluid separated from the lubricating oil by the purifier to the waste fluid tank; a sludge tank configured to store the waste fluid from the waste fluid tank; a waste fluid discharge line configured to supply the waste fluid in the waste fluid tank to the sludge tank; a neutralizer supply line configured to supply a neutralizer to the waste fluid tank; and a sensor configured to measure pH of the waste fluid or an ammonia concentration of the waste fluid in the waste fluid tank.
3. A neutralization processing system for an ammonia-containing waste fluid, the neutralization processing system separating sludge and ammonia water from lubricating oil after use in a ship diesel engine to neutralize a waste fluid containing at least sludge and ammonia water, the neutralization processing system comprising: a purifier configured to purify the lubricating oil; a waste fluid tank; a sludge discharge line configured to supply the sludge separated from the lubricating oil by the purifier to the waste fluid tank; a heavy fluid discharge line configured to supply the ammonia water separated from the lubricating oil by the purifier to the waste fluid tank; a neutralizer supply line configured to supply a neutralizer to the waste fluid tank; a sensor configured to measure pH of the waste fluid or an ammonia concentration of the waste fluid in the waste fluid tank; and a sludge tank configured to store the waste fluid from the waste fluid tank.
4. A neutralization processing system for an ammonia-containing waste fluid, the neutralization processing system separating sludge and ammonia water from lubricating oil after use in a ship diesel engine to neutralize a waste fluid containing at least sludge and ammonia water, the neutralization processing system comprising: a purifier configured to purify the lubricating oil; a sludge tank; a sludge discharge line configured to supply the sludge separated from the lubricating oil by the purifier to the sludge tank; a heavy fluid discharge line configured to supply the ammonia water separated from the lubricating oil by the purifier to the sludge tank; a neutralizer supply line configured to supply a neutralizer to the sludge tank; and a sensor configured to measure pH of the waste fluid or an ammonia concentration of the waste fluid.
5. The neutralization processing system for an ammonia-containing waste fluid according to claim 1, wherein the neutralizer has an acidic component.
6. The neutralization processing system for an ammonia-containing waste fluid according to claim 1, wherein the sensor is either or both of a pH meter and an ammonia sensor.
7. The neutralization processing system for an ammonia-containing waste fluid according to claim 1, wherein the neutralizer contains citric acid.
8. The neutralization processing system for an ammonia-containing waste fluid according to claim 7, further comprising: an additive for detoxification supply line configured to supply iron oxide to the sludge tank.
9. The neutralization processing system for an ammonia-containing waste fluid according to claim 2, further comprising: an additive for detoxification supply line configured to supply iron oxide to the waste fluid tank, wherein the neutralizer contains citric acid.
10. The neutralization processing system for an ammonia-containing waste fluid according to claim 1, wherein the sludge tank is provided with a device having a stirring function of stirring the waste fluid.
11. The neutralization processing system for an ammonia-containing waste fluid according to claim 2, wherein the waste fluid tank is provided with a device having a stirring function of stirring the waste fluid.
12. The neutralization processing system for an ammonia-containing waste fluid according to claim 1, wherein the sludge tank is provided with an input water supply line for supplying water to the sludge tank.

13. The neutralization processing system for an ammonia-containing waste fluid according to claim 2, wherein the waste fluid tank is provided with an input water supply line for supplying water to the waste fluid tank.

14. The neutralization processing system for an ammonia-containing waste fluid according to claim 2, further comprising: a liquid level measuring instrument provided with the waste fluid tank, the liquid level measuring instrument being configured to measure an amount of a neutralized waste fluid stored in the waste fluid tank; and a waste fluid transfer pump configured to transfer the waste fluid in the waste fluid tank to the sludge tank, wherein a discharge rate of the waste fluid transfer pump is controlled in accordance with the amount of the neutralized waste fluid measured by the liquid level measuring instrument.

15. A neutralization processing method for an ammonia-containing waste fluid, the neutralization processing method separating a waste fluid containing at least sludge and ammonia from lubricating oil after use in a ship diesel engine to neutralize the waste fluid, the neutralization processing method comprising: purifying the lubricating oil by a purifier; supply the sludge separated from the lubricating oil, by the purifier, to the waste fluid tank by a sludge discharge line; measuring pH or an ammonia concentration of the waste fluid in the waste fluid tank by a sensor; and neutralizing the waste fluid by adding a neutralizer to the waste fluid in accordance with either or both of a measured value of the pH of the waste fluid and a measured value of the ammonia concentration of the waste fluid measured by the sensor.
